

Homework Set 1 — Answers

Fall 2026

1. Calculation of Transition Probabilities

One-step transition probability matrix (states 1, 2, 3):

	1	2	3
1	0.4	0.5	0.1
2	0.3	0.3	0.4
3	0.3	0.2	0.5

(a) $P(X_2 = 2 \mid X_0 = 1) = 0.37$

(b) $P(X_0 = 2, X_1 = 3, X_2 = 1, X_3 = 2) = 0.03$

with $P(X_0 = 2) = 0.5$.

2. Credit Ratings

(a) Renormalized one-year transition probabilities (conditioned on remaining rated). Rows sum to 1.

From \ To	AAA	AA	A	BBB	BB	B	CCC/C	D
AAA	0.900537	0.092035	0.005262	0.000310	0.001032	0.000310	0.000516	0.000000
AA	0.004674	0.911300	0.077898	0.004570	0.000519	0.000623	0.000208	0.000208
A	0.000208	0.015425	0.931944	0.048359	0.002397	0.001042	0.000104	0.000521
BBB	0.000000	0.000742	0.032347	0.926185	0.034044	0.004242	0.000955	0.001485
BB	0.000110	0.000220	0.001102	0.048915	0.869120	0.068855	0.005508	0.006169
B	0.000000	0.000228	0.000685	0.001598	0.051033	0.858317	0.054687	0.033451
CCC/C	0.000000	0.000000	0.000824	0.001530	0.004708	0.155114	0.530423	0.307403

D is treated as absorbing: $p_{D,D} = 1$.

(b) $P(\text{BB to default within two years}) = 0.015601 (\approx 1.5601\%)$

(c) $P(\text{AAA to AAA or AA in three years}) = 0.958450 (\approx 95.8450\%)$

3. Gambler's Ruin

(a) State space $\{0, 1, \dots, N\}$. Transition probability matrix:

$p_{0,0} = 1$, $p_{N,N} = 1$; for $0 < i < N$: $p_{i,i+1} = p$, $p_{i,i-1} = 1 - p$, and $p_{i,j} = 0$ otherwise.

(b) No. States 0 and N are absorbing and do not communicate with other states.

(c) A stationary distribution exists but it is not unique.

The chain is finite; e.g. δ_0 , δ_N and any convex combination thereof are stationary.

4. (s, S)-Policy

State space $\{0, 1, \dots, S\}$. Let D denote generic one-period demand with $P(D = k) = \alpha_k$, $k = 0, 1, \dots$

For $i \geq s$:

$$p_{i,j} = \alpha_{i-j} \text{ for } 1 \leq j \leq i; \quad p_{i,0} = \sum_{k \geq i} \alpha_k; \quad p_{i,j} = 0 \text{ for } j > i.$$

For $i < s$ (order up to S):

$$p_{i,j} = \alpha_{s-j} \text{ for } 1 \leq j \leq S; \quad p_{i,0} = \sum_{k \geq S} \alpha_k; \quad \text{independent of } i.$$

5. PageRank

Out-degrees: $\deg(1) = 2$ (to 3, 5), $\deg(2) = 2$ (to 1, 3), $\deg(3) = 2$ (to 2, 4), $\deg(4) = 2$ (to 1, 5), $\deg(5) = 1$ (to 3). With damping $0.8 + \text{uniform } 0.2/5 = 0.04$:

	1	2	3	4	5
1	0.04	0.04	0.44	0.04	0.44
2	0.44	0.04	0.44	0.04	0.04
3	0.04	0.44	0.04	0.44	0.04
4	0.44	0.04	0.04	0.04	0.44
5	0.04	0.04	0.84	0.04	0.04

Stationary distribution $(\pi_1, \dots, \pi_5)$:

(0.173422, 0.166777, 0.316944, 0.166777, 0.176080).

6. Random Walk on a Graph

Irreducible: Yes. G connected implies any two nodes are joined by a positive-probability path.

Stationary distribution (for finite V ; otherwise the same invariant measure):

Let $w(x) = \sum_{z \in N_G(x)} w(\{x, z\})$ and $W = \sum_{x \in V} w(x)$. Then

$$\pi_x = w(x) / W, \quad x \in V.$$

End of answers.